Lab 1 – Requirements Engineering & UML Use-Case Modelling

#09 – Faculty Research Grant & Publication Tracker

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Comments
FR-001	Functional	The system shall track grant fund allocations, allow faculty to log equipment procurement expenses, and calculate the remaining grant budget.	High	Pass: Remaining balance updates accurately and expenses exceeding the grant cap are blocked. Fail: Overdraft is allowed without admin override.	Prevents grant overspending and provides accurate financial tracking.	Sample requirement from problem statement.
FR-002	Functional	The system shall allow faculty researchers to record and manage publication details, including title, authors, venue, and publication status.	High	Pass: Publication details are saved and can be retrieved or updated. Fail: Required publication information cannot be saved.	Maintains accurate faculty publication records for research tracking.	Supports publication management.

FR-003	Functional	The system shall track publication metrics including indexing status and citation information.	Medium	Pass: Indexing status and citation information are displayed correctly for each publication. Fail: Publication metrics cannot be retrieved or are incorrect.	Helps the research dean and faculty monitor publication impact and research performance.	Supports publication metrics tracking.
FR-004	Functional	The system shall allow faculty researchers to submit and manage co-author approval requests for publications.	Medium	Pass: Co-author approval requests can be submitted, reviewed, and their status is recorded. Fail: Approval status cannot be updated or retrieved.	Supports the co-author approval workflow for research publications.	Supports collaborative publication management.
FR-005	Functional	The system shall provide fund burn-up analytics to help the Research Dean monitor grant expenditure and remaining funds.	Medium	Pass: Burn-up analytics display total allocation, expenditure, and remaining funds correctly. Fail: The displayed financial values	Helps the Research Dean monitor research fund utilization and identify potential overspending.	Supports financial monitoring and decision-making.

				are missing or inaccurate.		
NFR-001	Nonfunctional	The system shall maintain an immutable, tamper-evident audit ledger for all financial approvals and journal status modifications.	High	Pass: Audit records cannot be modified or deleted without detection and all financial approvals are logged. Fail: An audit record can be altered without detection.	Ensures financial approvals and journal modifications remain secure and traceable.	Security and auditability requirement.
NFR-002	Nonfunctional	The system shall provide publication and grant information within 3 seconds under normal operating conditions. **4. Priority**	Medium	Pass: At least 95% of requests return the required information within 3 seconds during normal operation. Fail: More than 5% of requests exceed 3 seconds.	Provides responsive access to research and financial information for faculty and the Research Dean.	Performance requirement.